Assignment-3.

1) double number pyramid :

```
      1                    ___1
    1   2                  __1 2
  1   2   3      =        _1 2 3
1 2   3   4                1 2 3 4
  1 2 3
    1 2
      1
```

first spaces are decreased so $(n-i)$ = space.
take variable inside outer for loop.
first inner loop is for space.
and second for print values.

```
n=4
for (int i=1; i<=n; i++) {
        int count = 1;
        for (int j=1; j<=n-i; j++) {
                System.out.print(" ");
        }
        for (int j=1; j<=i; j++) {
                System.out.print(count + " ");
        }
        System.out.println();
}
// for inverted pyramid. - reverse outer loop.

for (int i=n-1; i>=1; i--) {
        ~~for~~ int count = 1;
        for (int j=1; j<=n-i; j++) {
                Sop(" ");
        }
```

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```
for (int j=1; j<=i ; j++){
        Sop ( count+ " ");
    }
    System.out.println ( );
}
```

Dryrun : $i=1$ $1<=4$ $n-i=3$

└ $j=1 <=3$ — space.

$j=2 <=3$ — space.

$j=3 <=3$ — space.

$j=4 <=3$ — loop terminate.

Second loop = : $i=1$

$j=1 <=1$ — print count. $=1$

$j=2 <=1$ — loop terminate.

$i=2$ $2<=4$ $n-i = 4-2 = 2$

└ $j=1<=2$ — space.

$j=2<=2$ — space.

$j=3<=2$ — X stop.

Second loop =

$j=1<=2$ — Count $=1$

$j=2<=2$ — 2.

$j=3<=2$ — X stop.

$i=3$ $3<=4$ $n-i = 4-3 = 1$

└ $j=1<=1$ = space.

$j=2<=1$ — X stop.

Second loop =

$j=1<=3$ — 1

$j=2<=3$ — 2

$j=3<=3$ — 3

$j=4<=3$ — X stop.

$i=4$ $4<=4$ $n-i = 4-4 = 0$.

└ $j=1<=0$ — X stop.

Second loop :

$j=1<=4 = 1$

$j=2<=4 = 2$

j=3 <= 4 — 3
j=4 <= 4 — 4
j=5 <= 4 — × stop.

first outer loop is Completed.
O/P is:

```
1
1 2
1 2 3
1 2 3 4.
```

next inverted pyramid.

i = n-1 = 3 3>=1 n-i = 4-3 = 1
└ j=1 <= 1 — space.
j=2 <= 1 — × stop.

second loop:
j=1 <= 3 — 1
j=2 <= 3 — 2
j=3 <= 3 — 3

i = 2 2>=1 4-2 = 2
└ j=1 <= 2 = space.
j=2 <= 2 - space.

Second loop:
j=1 <= 2 - 1
j=2 <= 2 - 2.

i=1 1>=1 n-i = 3
└ j=1 <= 3 - space.
j=2 <= 3 - space.
j=3 <= 3 - space.

Second loop :

j=1 <=1 — 1

j=2 <=1 — X Stop.

Second outer loop Completed.

Output is :

```
    1
   1 2
  1 2 3
 1 2 3 4
  1 2 3
   1 2
    1
```

Q.2. number butterfly. :

n = 4

```
i=
1   1_______1
2   12_____21
3   123___321
4   1234 4321
5   1234 4321
6   123   321
7   12     21
8   1       1
```

Space = 6
Space = 4
Space : 2
Space = 2 × n - i

```
1                 1
12              21
123           321
1234 4321
```

= first print number then space then number.

n = 4

```
for (int i=1 ; i<=n ; i++){
    for (int j=1; j<=i ; j++){
        System.out.print (j);
    }
    for (int j=1 ; j<=2*(n-i) ; j++){
        System.out.print (" ");
    }
    for (int j=i ; j>=1 ; j--){
        System.out. print (j);
    }
    System.out. println( );
}
```

```
for (int i=n ; i>=1 ; i--) {
    for (int j=1 ; j<=i ; j++) {
            System.out.print (j);
    }
    for (int j=1 ; j<=2*(n-i) ; j++) {
            System.out.print (" ");
    }
    for (int j=i ; j>=1 ; j--) {
            System.out.print (j);
    }
    System.out.println ();
}
```

Dryrun :

i=1 <=4

j=1<=1 — 1

j=2<=1 — X

2*(n-i)=6

j=1<=6 — Space

j=2<=6 — space.

j=3<=6 — space.

j=4<=6 — Space.

j=5<=6 — Space.

j=6<=6 — Space.

j=7<=6 — X.

j=i=1 >=1 — 1

j=0 >=1 — X

$i = 2 <= 4$

$\quad$ j = 1 <= 2 — 1

$\quad$ j = 2 <= 2 — 2

$\quad$ j = 3 <= 2 — X

$2 \times (n-i) = 4$

$\quad$ j = 1 <= 4 — Space.

$\quad$ j = 2 <= 4 — space.

$\quad$ j = 3 <= 4 — Space.

$\quad$ j = 4 <= 4 — Space.

$\quad$ j = 5 <= 4 — X,

j = i = 2 >= 1 — 2

$\quad$ 1 >= 1 — 1

$\quad$ 0 >= 1 — X.

$i = 3 <= 4$

$\quad$ j = 1 <= 3 — 1

$\quad$ j = 2 <= 3 — 2

$\quad$ j = 3 <= 3 — 3

$2 \times (n-i)$ — 2

$\quad$ j = 1 <= 2 — Space.

$\quad$ j = 2 <= 2 — Space.

j = i = 3 >= 1 — 3

$\quad$ 2 >= 1 — 2

$\quad$ 1 >= 1 — 1

$i = 4 <= 4$

$\quad$ j = 1 <= 4 — 1

$\quad$ j = 2 <= 4 — 2

$\quad$ 3 <= 4 — 3

$\quad$ 4 <= 4 — 4

$2 \times (n-i) - 2 \times (4-4) = 0$.

$j = 1 < 0 \rightarrow$.

$j = i = 4$

$j = 4 >= 1 - 4$

$j = 3 >= 1 - 3$

$j = 2 >= 1 - 2$

$j = 1 >= 1 - 1$

first loop output is :

```
1          1
12        21
123      321
1234 4321
```

Second loop :

$i = 4 \; ; \; 4 >= 1$

$\llcorner$ $j = 1 <= 4 - 1$

$j = 2 <= 4 - 2$

$3 <= 4 - 3$

$4 <= 4 - 4$

$2(n-i) = 0$ - no space.

$j = i = 4$

$j = 4 >= 1 - 4$

$j = 3 >= 1 - 3$

$2 >= 1 - 2$

$1 >= 1 - 1$

$i = 3 \quad 3 > 1$

$\llcorner$ $j = 1 <= 3 - 1$

$2 <= 3 - 2$

$3 <= 3 - 3$

$2(n-i) = 2$

$j = 1 <= 2$ - space.

$j = 2 <= 2$ - space.

$j = i = 3$

$3 >= 1$ — 3

$2 >= 1$ — 2

$1 >= 1$ — 1

$i = 2 >= 1$

$j = 1 <= 2$ — 1

$j = 2 <= 2$ — 2

$2(n-i) = 4$

$j = 1 <= 4$ — space.

$2 <= 4$ — space.

$3 <= 4$ — space.

$4 <= 4$ — space.

$j = i = 2$

$2 >= 1$ — 2

$1 >= 1$ — 1

$i = 1 >= 1$

$j = 1 <= 1$ — 1

$2(n-i) = 6$

$j = 1 <= 6$ — Space.

$2 <= 6$ — space.

$3 <= 6$ — Space

$4 <= 6$ — Space

$5 <= 6$ — space

$6 <= 6$ — Space.

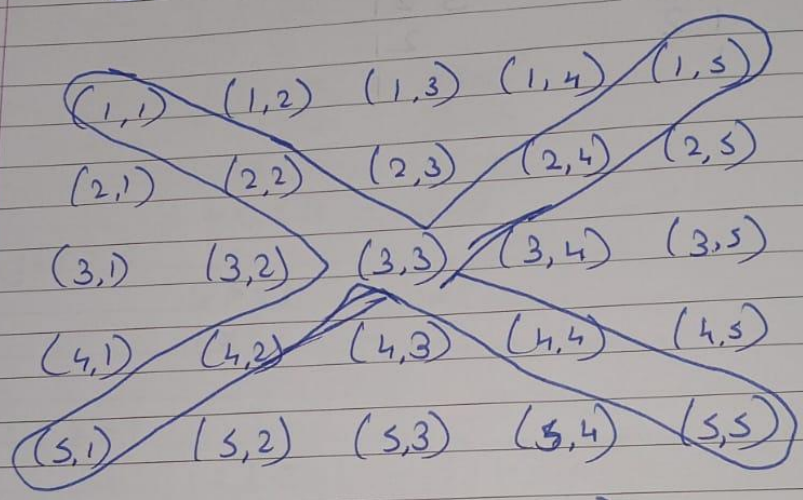
$j = i = 1$

$1 >= 1$ — 1

final o/p is:

```
1                 1
1 2             2 1
1 2 3         3 2 1
1 2 3 4     4 3 2 1
1 2 3 4     4 3 2 1
1 2 3         3 2 1
1 2             2 1
1                 1
```

Q3 Cross (x) $n=5$



(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)

```
if (i==j || i+j==n+1)
        print (i)

for (int i=1; i<=n; i++){
    for (int j=1; j<=n; j++){
        if (i==j || i+j==n+1) {
            System.out.print (i);
        }else{
            System.out.print (" ");
        }
    }
    Sop ();
}
```

Dryrun : i = 1 <= 5

└ j = 1 <= 5

└ i == j - true. 1

j = 2 <= 5

└ i == j - X

└ j == 6 X. - Space.

j = 3 <= 5

└ i == j - X

└ 4 == 6 - X

j = 4 <= 5

└ i == j - X

└ 5 == 6 - X

j = 5 <= 5

└ i == j - X

└ 6 == 6 - ✓ 1

i = 2 <= 5 i == j X

└ j = 1 <= 5

└ 3 == 6 X

j = 2 <= 5

└ i == j - 2

j = 3 <= 5

└ 5 == 6 - X.

j = 4 <= 5

└ 6 == 6 - 2

j = 5 <= 5

└ 7 == 6 - X.

i = 3 <= 5

└ j = 1 <= 5

└ 4 == 6 - X

j=2<=5
└ 5==6 ✗
j=3<=5
└ 8==3 — 3
j=4<=5
└ 7==6 - ✗
j=5<=5
└ 8==6 ✗.

i=4 <=5
└ j=1<=5
└ 5==6 ✗
j=2<=5
└ 6==6 ✓ 4
j=3<=5
└ 7==6 ✗
j=4<=5
└ 4==4 — ✓ 4
j=5<=5
└ 9==6 ✗.

i=5 <=5
└ j=1<=5
└ 6==6 — ✓ 5
j=2<=5
└ 7==6 — ✗ (i+j==n+1)
j=3<=5
└ 8==6 — ✗
j=4<=5
└ 9==6 — ✗
j=5<=5
└ 5==5 — ✓ 5

Output is :

```
1       1
  2   2
    3
  4   4
5       5
```

Q.5

```
      1
    2   3
  4   5   6
7   8   9   10
  11  12  13
    14  15
      16 .
```

declare count var at top.

```
int n = 4;
int count = 1;
for (int i = 1; i<=n; i++){
        for (int j = 1; j<= n-i; j++){
                System.out.print (" ");
        }
        for (int j = 1; j<= i; j++){
                System.out.print (count + " ");
                Count ++;
        }
        ~~for (int j = 1; j~~
    }
    System.out.println ();
}
```

```
for (int i = n-1; i>=1; i--){
        for (int j = 1; j<= n-i; j++){
                System.out.print (" ");
        }
        for (int j = 1; j<= i; j++){
                System.out.print (count + " ");
                count ++;
        }
        System.out.println ();
}
```

Output :

```
      1
    2   3
  4   5   6
7   8   9   10
  11  12  13
    14  15
      16
```

Q.7

```
      1
    0   1
  1   0   1
0   1   0   1
  1   0   1
    0   1
      1
```

(1+1)=2	(1,2)	(1,3)	(1,4)
(2+1)=3	(2,2) 4	(2,3)	(2,4)
(3+1)=4	(3+2)=5	(3,3)	(3,4)
(4,1)	(4,2)	(4,3)	(4,4)

if (i+j %2==0) then print 1 otherwise 0.

```
for (int i=1; i<=n; i++){
        for (int j=1; j<=n-i; j++){
                System.out.print(" ");
        }
        for (int j=1; j<=i; j++){
                int total = i+j;
                if (total %2==0){
                        System.out.print(1+" ");
                }else {
                        System.out.print(0+" ");
                }
        }
        System.out.println();
}
for (int i=n-1; i>=1; i--){
        for (int j=1; j<=n-i; j++){
                system.out.print(" ");
        }
        for (int j=1; j<=i; j++){
                int total = i+j;
                if (total %2==0){
                        sop(1+" ");
}
```

```
else {
        Sop (0 + " ");
    }
}
System.out.println();
}
```

O/P :-

```
        1
      0   1
    1   0   1
  0   1   0   1
    1   0   1
      0   1
        1
```